Janith Ramanayake

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Personal Profile

A highly motivated Data Science graduate with a solid academic foundation in statistics, mathematics, and computer science, complemented by hands-on research experience in machine learning and deep learning. I am currently seeking PhD opportunities in Statistics, with research interests in applying statistical and computational methods to multidisciplinary problems. Passionate about addressing real-world problems through data-driven, interdisciplinary approaches, and dedicated to advancing knowledge through collaborative research and impactful scientific contributions.

Education

BSc (Hons) in Data Science Apr 2021 – Apr 2025
University of Colombo | Faculty of Science GPA : 3.87|4 (First Class)

- **Statistics:** Probability and Distributions, Statistical Theory, Statistical Inference I/II, Multivariate Data Analysis, Time Series Analysis, Computational Statistics, Applied Statistical Models
- **Computer Science:** Introduction to Data Structures and Algorithms, Machine Learning I/II, Deep Learning, DataBase Systems I/II, Natural Language Processing, Image Analysis, Big Data Analytics, Fundamentals of Programming, Internet Technologies
- **Applied Mathematics:** Differential Equations I/II, Applied Linear Algebra, Computational Mathematics I/II, Linear Programming, Numerical Analysis, Optimization
- **Pure Mathematics:** Basic Analysis I/II/III, Introduction to Abstract Algebra, Introduction to Number Theory
- **Undergraduate Thesis:** Enhancing Sleep Stage Classification with Vision Transformers: A Study on Image Transformations, Data Augmentation, and Model Optimization

Certificate in Quantum Computing Jan 2026 – Jun 2026
University of Colombo | Faculty of Science & Centre for Mathematical Modelling Final Result: Honours

- **Topics:** Quantum computing fundamentals, quantum circuits, quantum algorithms (Deutsch–Jozsa, Grover's, Shor's), Quantum Fourier Transform (QFT), Qiskit, SpinQ, quantum annealing, QUBO, and quantum applications.
- **Research Project:** Developed a toy implementation of Quantum Fourier Transform (QFT)-based image compression using Qiskit to demonstrate quantum image processing concepts.

GCE Advanced Level - Physical Science Aug 2019
Mahinda College | Galle BBB

Work Experience

Research Assistant Jul 2026 – Present
Department of Computer Science & Engineering - University of Moratuwa Moratuwa, LK

- Conducting research on the socio-cultural alignment of large language models for the Sinhala language.

Research Assistant (Part-time) Jun 2026 – Present
Department of Statistics and Computer Science - University of Peradeniya Peradeniya, LK

- Conducting research on improving cybersecurity awareness through gamification techniques.

Visiting Lecturer Jan 2026 – Jun 2026
Department of Statistics - University of Colombo Colombo, LK

- Conducted lectures for the DA 2002 - Operating Systems course in the BSc (External) in Applied Data Analytics degree programme.

Temporary Assistant Lecturer May 2025 – Jun 2026

Department of Statistics - University of Colombo Colombo, LK

- Conducted lectures, practical laboratory, and tutorial sessions for the following Level 1 and Level 2 undergraduate courses:
  - IS 1007 - Introduction to Statistical Computing
  - ST 1008 - Probability & Distributions
  - ST 1009 - Exploratory Data Analysis
  - ST 2007 - Applications in Statistical Inference

Intern Data Scientist Nov 2024 – Apr 2025
Central Bank of Sri Lanka Colombo, LK

- Collaborated with senior statisticians and economists to integrate alternative data sources into macroeconomic analysis and policy research.
- Contributed to research and development of data-driven economic indicators using alternative data sources and machine learning techniques.
- Supported data visualization and reporting efforts by creating clear, informative charts and dashboards to communicate analytical insights to non-technical stakeholders.

Research Experience

Enhancing Sleep Stage Classification with Vision Transformers: A Study on Image Transformations, Data Augmentation, and Model Optimization Jun 2024 - Mar 2025
Undergraduate Thesis, University of Colombo

- Investigated signal-to-image transformation techniques (spectrograms, Gramian angular fields) to convert EEG signals into visual representations for sleep stage classification, applied GAN-based synthetic image generation to address class imbalance and improve subject-to-subject invariance, and benchmarked fine-tuned pre-trained Vision Transformer models (ViT-B/16, ViT-B/32, DeiT-B, Swin-B, PVT-Small) against conventional deep learning approaches.

A Data Driven Approach to Measuring Consumer Confidence in Sri Lanka with Google Search Trends and Deep Learning Jan 2025 - Mar 2025
Central Bank of Sri Lanka

- Developed a real-time Consumer Confidence Index for Sri Lanka by collecting and engineering high-frequency Google Search Trends features related to economic sentiment, consumption, and employment, training deep learning models (LSTM, GRU, and hybrid architectures), and validating index trends against historical economic indicators and major socio-economic events.

Cross-Lingual Sentiment Modeling for Financial Texts in English, Sinhala, and Tamil Nov 2024 - Jan 2025
Central Bank of Sri Lanka

- Curated and annotated a trilingual financial text dataset from news articles, press releases, and social media, fine-tuned the Llama 3.2 1B large language model for domain-adapted sentiment classification, and addressed low-resource challenges in Sinhala and Tamil through translation-based augmentation, evaluating performance across languages using precision, recall, and F1-score.

Current Personal Projects

Physics-Informed Spatial Deep Learning for Dengue Prediction

- Developing a physics-informed spatial deep learning framework that integrates satellite-derived environmental data, hotspot-guided spatial feature engineering, and Ross–Macdonald disease transmission dynamics for regional dengue prediction across Sri Lanka.

BiCQNet: Bidirectional Classical–Quantum Neural Network for Image Classification

- Developing a hybrid classical–quantum deep learning architecture that enables bidirectional interaction between CNN feature representations and parameterized quantum circuits through multi-stage quantum–classical feature fusion for enhanced image classification.

Publications

Systematic Evaluation of Signal-to-Image Transformation Pipelines for Vision Transformer-based Sleep Stage Classification **Under Review**
Conference Paper - SCICASH 2026

Quantum Mutual Information for Causal Feature Selection: An Empirical Study of Feature Recovery and Treatment Effect Estimation **Under Review**
Conference Paper - ICTer 2026

Ramanayake, J. R. R. H., Arachchige, C. N. P. G., (2025). ***Representation Separability of Signal-to-Image Transformations for Sleep Stage Classification Using Pretrained Vision Transformers.***
Oral presentation at the 3rd International Conference in Data Science (ICDS 2025)

Abilash, S., Perera, A. S., **Ramanayake, J. R. R. H.**, Amarathunga, J., Nagodawithana, K. T., (2026). ***Machine Learning Framework for Predicting Thermophysical Properties of Polymers from SMILES Representations.***
Oral presentation at the 13th Ruhuna International Science and Technology Conference (RISTC 2026)

Technologies

Programming Languages: Python, R, Java, JavaScript

Mathematics and Statistics: Octave, SPSS, Minitab, Microsoft Excel

Frontend and Backend Development: React, React Native, Flask, FastAPI

Machine Learning and Data Science: TensorFlow, PyTorch, Power BI

Database Management: MySQL, MongoDB, BigQuery

Awards and Achievements

Honored on the Dean's List for academic excellence	2021 - 2024
Awarded Faculty of Science Colours for Table Tennis	2023 - 2024
Finalist, Intellihack NextGen Generative AI Hackathon organized by UCSC	2024
Finalist, MiniHack Machine Learning Hackathon organized by Stat Circle, UoC FoS	2023
Awarded Junior National Statistics Olympiad Merit Award.	2015 - 2016

Extra Curricular Activities

Member of Stat Circle, Department of Statistics, University of Colombo	May 2023 - Apr 2025
Member of Table Tennis Club, University of Colombo	May 2021 - Apr 2024
Member of Leo Club, Faculty of Science, University of Colombo	May 2021 - Apr 2023

References

Dr. G A C N Priyadarshani
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Department of Statistics
University of Colombo
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Dr. H A S G Dharmarathne
Senior Lecturer
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